blood

722. CLINICAL ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD, IMMUNE RECONSTITUTION: BIOMARKERS AND THE MICROBIOME | NOVEMBER 29, 2018

Intestinal Enterococcus Is a Major Risk Factor for the Development of Acute Gvhd

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Abstract

Introduction:

Increasing evidence suggests that the intestinal microbiota is involved in the development of acute graft-vs.-host disease (GVHD) after allogeneic hematopoietic cell transplantation (allo-HCT). We previously reported in single center studies that Enterococcus a) is associated with GVHD (Holler et al., BBMT 2014)

and b) can dominate the post-transplant gut microbiota in up to 50% of allo-HCT patients resulting in a 9-fold increased risk of bacteremia (Taur et al, CID 2012). To further investigate the hypothesis that Enterococcus can trigger the development of GVHD, we studied both allo-HCT patients and pre-clinical mouse transplant models.

Methods and Results:

Stool samples from 1240 allo-HCT patients at 4 different transplant centers in the U.S., Germany and Japan were collected approximately weekly during inpatient hospitalization. The V4-V5 region of the bacterial 16S rRNA genes from 6718 samples was sequenced at one central site on the Illumina platform. We observed Enterococcus mono-domination (relative abundance > 30%) in post-transplant samples ranging from 20 to 60% of patients at different centers (Fig. A, left). This mono-domination was primarily attributable to *E. faecium*, and was associated with a significantly increased risk for grade 2-4 acute GVHD (Fig. A, right). In three different mouse models we found a transient bloom of *E. faecalis* around 7 days after transplant in allo-HCT recipients with GVHD (Fig. B; C57BL/6 -> 129SV model). This bloom did not occur in allo-HCT recipients of a T cell depleted allograft without GVHD. To further investigate this Enterococcus bloom, we treated mice with an experimental E. faecalis-strain on days 4 to 6 after transplant and found significantly increased lethal GVHD. Colonizing germ-free mice with a minimal gut flora also lead to increased lethal GVHD when enterococci were added to the gnotobiotic flora (Fig. C). The allo-HCT recipients with Enterococcus-containing flora had also increased serum IFNg levels. Short chain fatty acids (SCFA) can be protective against GVHD and gut inflammation through maintenance of epithelial homeostasis and increases in anti-inflammatory regulatory T cells in the gut. In BMT mouse models, we found that Enterococcus-dominated allo-HCT recipients with GVHD have significantly less cecal butyrate, a major SCFA. Similarly, Enterococcus domination after allo-HCT also leads to a decrease in fecal SCFAs in patients (Fig. D). Next, we hypothesized that intestinal IgA might have a protective role against this pathogen in mice. 16S sequencing of flow sorted IgA-coated vs. non-coated bacteria from fecal samples of allo-HCT patients and transplanted mice revealed no specific IgA-coating pattern of enterococci both before or after transplant rather excluding the hypothesis that IgA might have a protective role against Enterococci. *E. faecalis* and *E. faecium* use the disaccharide lactose as a major carbohydrate source for growth and expansion as observed by analyses of the Enterococcus genome and in vitro growth experiments. In mice, we observed that a lactose-free diet significantly decreases the Enterococcus bloom after transplant in allo-T cell recipients and in first survival experiments attenuates lethal GVHD (Fig. E).

Conclusion:

Our studies in mouse and man demonstrate that the abundance of Enterococcus in the intestinal flora plays a role in the development of GVHD and the prevention of Enterococcus growth with a lactose-free diet can ameliorate GVHD.

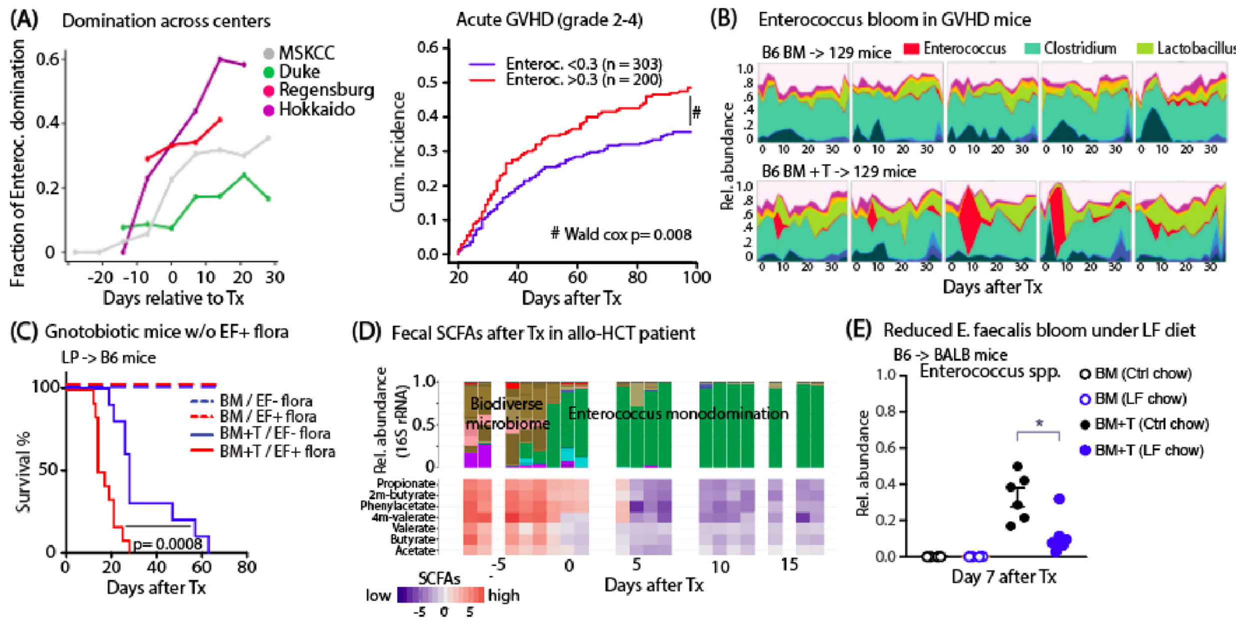

Figure. **(A)** *Left,* fraction of fecal samples with Enterococcus mono-domination of gut microbiota (rel. abundance >0.3) after allo-HCT (Tx) occurring at different transplant centers (MSKCC: 1030 pat., Duke: 71 pat., Regensburg: 81 pat., Hokkaido: 58 pat.). *Right,* significantly increased cumulative incidence of acute GVHD (grade 2-4, up to day 100) in patients at MSKCC that are Enterococcus mono-dominated (day 7 to 21 after allo-HCT). **(B)** High-density 16S rRNA gut microbiota sequencing of 129SV mice (1 box = 1 mouse) receiving transplantation of C57BL/6 bone marrow (BM) or BM plus T cells (4 Mio T cells; BM+T) developing acute GVHD. BM+T-transplanted mice displayed Enterococcus domination of the mouse gut microbiota around day 7 after transplant (Tx). **(C)** Germ-free C57BL/6 mice were colonized with minimal flora (Lactobacillus gasseri, Parabacteroides diastonis, Blautia producta, Bacteroides sartorii, Clostridium bolteae) and one group of mice was spiked with E. faecalis OG1RF before Tx with LP BM +/- T cells. Gnotobiotic mice harboring intestinal enterococci displayed significantly increased lethal acute GVHD. **(D)** Example of a patient with post-transplant Enterococcus mono-domination of the gut microbiota showing a concomitant reduction of SCFA, primarily butyrate, in fecal samples after allo-HCT. **(E)** Maintaining BALB/c mice on a lactose-free diet (day -7 to day 14 after Tx) significantly reduces Enterococcus rel. abundance at day 7 (16S rRNA qPCR) in BALB/c mice receiving C57BL/6 BM + T cells (0.5 Mio). $*p<0.05$.

Disclosures

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Author notes

* Asterisk with author names denotes non-ASH members.

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